Implementing Scenario 4

You can use the VPC wizard to create the VPC, subnets, NAT gateway, and optionally, an egress-only Internet gateway. You must specify an Elastic IP address for your NAT gateway; if you don't have one, you must first allocate one to your account. If you want to use an existing Elastic IP address, ensure that it's not currently associated with another instance or network interface. The NAT gateway is automatically created in the public subnet of your VPC.

These procedures include optional steps for enabling and configuring IPv6 communication for your VPC. You do not have to perform these steps if you do not want to use IPv6 in your VPC.

(Optional) To allocate an Elastic IP address for the NAT gateway (IPv4)

1. Open the Amazon VPC console at https://console.aws.amazon.com/vpc/.
2. In the navigation pane, choose **Elastic IPs**.
3. Choose **Allocate new address**.
4. Choose **Allocate**.

> **Note**
>
> If your account supports EC2-Classic, first choose **VPC**.

To create a VPC

1. Open the Amazon VPC console at https://console.aws.amazon.com/vpc/.
2. On the VPC dashboard, choose **Start VPC Wizard**.
3. Choose the second option, **VPC with Public and Private Subnets**, and **Select**.
4. For **VPC name**, **Public subnet name** and **Private subnet name**, you can name your VPC and subnets to help you identify them later in the console. You can specify your own IPv4 CIDR block range for the VPC and subnets, or you can leave the default values.
5. (Optional, IPv6-only) For **IPv6 CIDR block**, choose **Amazon-provided IPv6 CIDR block**. For **Public subnet's IPv6 CIDR**, choose **Specify a custom IPv6 CIDR** and specify the hexadecimal pair value for your subnet, or leave the default

value. For **Private subnet's IPv6 CIDR**, choose **Specify a custom IPv6 CIDR**. Specify the hexadecimal pair value for the IPv6 subnet or leave the default value.

6. In the **Specify the details of your NAT gateway** section, specify the allocation ID for an Elastic IP address in your account.
7. You can leave the rest of the default values on the page, and choose **Create VPC**.

Because the WebServerSG and DBServerSG security groups reference each other, create all the security groups required for this scenario before you add rules to them.

To create the WebServerSG and DBServerSG security groups

1. Open the Amazon VPC console at https://console.aws.amazon.com/vpc/.
2. In the navigation pane, choose **Security Groups**, **Create Security Group**.
3. Provide a name and description for the security group. In this topic, the name `WebServerSG` is used as an example. For **VPC**, select the ID of the VPC you created and choose **Yes, Create**.
4. Choose **Create Security Group** again.
5. Provide a name and description for the security group. In this topic, the name `DBServerSG` is used as an example. For **VPC**, select the ID of your VPC and choose **Yes, Create**.

To add rules to the WebServerSG security group

1. Select the WebServerSG security group that you created. The details pane displays the details for the security group, plus tabs for working with its inbound and outbound rules.
2. On the **Inbound Rules** tab, choose **Edit** and add rules for inbound traffic as follows:
    a. Choose **Type**, **HTTP**. For **Source**, enter `0.0.0.0/0`.
    b. Choose **Add another rule**, **Type**, **HTTPS**. For **Source**, enter `0.0.0.0/0`.
    c. Choose **Add another rule**, **Type**, **SSH**. For **Source**, enter your network's public IPv4 address range.
    d. Choose **Add another rule**, **Type**, **RDP**. For **Source**, enter your network's public IPv4 address range.
    e. (Optional, IPv6-only) Choose **Add another rule**, **Type**, **HTTP**. For **Source**, enter `::/0`.

    f. (Optional, IPv6-only) Choose **Add another rule**, **Type**, **HTTPS**. For **Source**, enter `::/0`.
    g. (Optional, IPv6-only) Choose **Add another rule**, **Type**, **SSH** (for Linux) or **RDP** (for Windows). For **Source**, enter your network's IPv6 address range.
    h. Choose **Save**.
3. On the **Outbound Rules** tab, choose **Edit** and add rules for outbound traffic as follows:
    a. Locate the default rule that enables all outbound traffic and choose **Remove**.
    b. Choose **Type**, **MS SQL**. For **Destination**, specify the ID of the DBServerSG security group.
    c. Choose **Add another rule**, **Type**, **MySQL**. For **Destination**, specify the ID of the DBServerSG security group.
    d. Choose **Add another rule**, **Type**, **HTTPS**. For **Destination**, enter `0.0.0.0/0`.
    e. Choose **Add another rule**, **Type**, **HTTP**. For **Destination**, enter `0.0.0.0/0`.
    f. (Optional, IPv6-only) Choose **Add another rule**, **Type**, **HTTPS**. For **Destination**, enter `::/0`.
    g. (Optional, IPv6-only) Choose **Add another rule**, **Type**, **HTTP**. For **Destination**, enter `::/0`.
    h. Choose **Save**.

To add the recommended rules to the DBServerSG security group

1. Select the DBServerSG security group that you created. The details pane displays the details for the security group, plus tabs for working with its inbound and outbound rules.
2. On the **Inbound Rules** tab, choose **Edit** and add rules for inbound traffic as follows:
    a. Choose **Type**, **MS SQL**. For **Source**, specify the ID of your WebServerSG security group.
    b. Choose **Add another rule**, **Type**, **MYSQL**. For **Source**, specify the ID of your WebServerSG security group.
    c. Choose **Save**.

3. On the **Outbound Rules** tab, choose **Edit** and add rules for outbound traffic as follows:
    a. Locate the default rule that enables all outbound traffic and choose **Remove**.
    b. Choose **Type**, **HTTP**. For **Destination**, enter `0.0.0.0/0`.
    c. Choose **Add another rule**, **Type**, **HTTPS**. For **Destination**, enter `0.0.0.0/0`.
    d. (Optional, IPv6-only) Choose **Add another rule**, **Type**, **HTTP**. For **Destination**, enter `::/0`.
    e. (Optional, IPv6-only) Choose **Add another rule**, **Type**, **HTTPS**. For **Destination**, enter `::/0`.
    f. Choose **Save**.

You can now launch instances into your VPC.

To launch an instance (web server or database server)

1. Open the Amazon EC2 console at https://console.aws.amazon.com/ec2/.
2. From the dashboard, choose **Launch Instance**.
3. Select an AMI and an instance type and choose **Next: Configure Instance Details**.

    Note

    If you intend to use your instance for IPv6 communication, you must choose a supported instance type; for example, T2. For more information, see Amazon EC2 Instance Types.

4. On the **Configure Instance Details** page, for **Network**, select the VPC that you created earlier and then select a subnet. For example, launch a web server into the public subnet and the database server into the private subnet.
5. (Optional) By default, instances launched into a nondefault VPC are not assigned a public IPv4 address. To be able to connect to your instance in the public subnet, you can assign a public IPv4 address now, or allocate an Elastic IP address and assign it to your instance after it's launched. To assign a public IPv4 address now, ensure that you choose **Enable** from the **Auto-assign Public**

IP list. You do not need to assign a public IP address to an instance in the private subnet.

> **Note**
>
> You can only use the auto-assign public IPv4 feature for a single, new network interface with the device index of eth0. For more information, see Assigning a Public IPv4 Address During Instance Launch.

6. (Optional, IPv6-only) You can auto-assign an IPv6 address to your instance from the subnet range. For **Auto-assign IPv6 IP**, choose **Enable**.
7. On the next two pages of the wizard, you can configure storage for your instance, and add tags. On the **Configure Security Group** page, choose the **Select an existing security group** option, and select one of the security groups you created earlier (**WebServerSG** for a web server or **DBServerSG** for a database server). Choose **Review and Launch**.
8. Review the settings that you've chosen. Make any changes that you need and choose **Launch** to choose a key pair and launch your instance.

If you did not assign a public IPv4 address to your instance in the public subnet in step 5, you will not be able to connect to it. Before you can access an instance in your public subnet, you must assign it an Elastic IP address.

To allocate an Elastic IP address and assign it to an instance (IPv4)

1. Open the Amazon VPC console at https://console.aws.amazon.com/vpc/.
2. In the navigation pane, choose **Elastic IPs**.
3. Choose **Allocate new address**.
4. Choose **Allocate**.

> **Note**
>
> If your account supports EC2-Classic, first choose **VPC**.

5. Select the Elastic IP address from the list and choose **Actions**, **Associate address**.
6. Select the network interface or instance. For **Private IP**, select the corresponding address to associate the Elastic IP address with and choose**Associate**.

You can now connect to your instances in the VPC. For information about how to connect to a Linux instance, see Connect to Your Linux Instance in the *Amazon EC2 User Guide for Linux Instances*. For information about how to connect to a Windows instance, see Connect to Your Windows Instance in the*Amazon EC2 User Guide for Windows Instances*.

Implementing Scenario 2 with a NAT Instance

You can implement scenario 2 using a NAT instance instead of a NAT gateway. For more information about NAT instances, see NAT Instances.

You can follow the same procedures as above; however, in the NAT section of the VPC wizard, choose **Use a NAT instance instead** and specify the details for your NAT instance. You will also require a security group for your NAT instance (`NATSG`), which allows the NAT instance to receive Internet-bound traffic from instances in the private subnet, as well as SSH traffic from your network. The NAT instance can also send traffic to the Internet, so that instances in the private subnet can get software updates.

After you've created the VPC with the NAT instance, you must change the security group associated with the NAT instance to the new `NATSG` security group (by default, the NAT instance is launched using the default security group).

NATSG: Recommended Rules

Inbound			
Source	**Protocol**	**Port Range**	**Comments**
10.0.1.0/24	TCP	80	Allow inbound HTTP traffic from database servers that are in the private subnet
10.0.1.0/24	TCP	443	Allow inbound HTTPS traffic from database servers that are in the private subnet
Your network's public IP address range	TCP	22	Allow inbound SSH access to the NAT instance from your network (over the Internet gateway)
Outbound			

Destination	Protocol	Port Range	Comments
0.0.0.0/0	TCP	80	Allow outbound HTTP access to the Internet (over the Internet gateway)
0.0.0.0/0	TCP	443	Allow outbound HTTPS access to the Internet (over the Internet gateway)

To create the NATSG security group

1. Open the Amazon VPC console at https://console.aws.amazon.com/vpc/.
2. In the navigation pane, choose **Security Groups**, and the choose **Create Security Group**.
3. Specify a name and description for the security group. In this topic, the name `NATSG` is used as an example. For **VPC**, select the ID of your VPC and choose **Yes, Create**.
4. Select the NATSG security group that you created. The details pane displays the details for the security group, plus tabs for working with its inbound and outbound rules.
5. On the **Inbound Rules** tab, choose **Edit** and add rules for inbound traffic as follows:
    a. Choose **Type**, **HTTP** . For **Source**, enter the IP address range of your private subnet.
    b. Choose **Add another rule**, **Type**, **HTTPS**. For **Source**, enter the IP address range of your private subnet.
    c. Choose **Add another rule**, **Type**, **SSH**. For **Source**, enter your network's public IP address range.
    d. Choose **Save**.
6. On the **Outbound Rules** tab, choose **Edit** and add rules for outbound traffic as follows:
    a. Locate the default rule that enables all outbound traffic and choose **Remove**.
    b. Choose **Type**, **HTTP**. For **Destination**, enter `0.0.0.0/0`.
    c. Choose **Add another rule**, **Type**, **HTTPS**. For **Destination**, enter `0.0.0.0/0`.
    d. Choose **Save**.

When the VPC wizard launched the NAT instance, it used the default security group for the VPC. You need to associate the NAT instance with the NATSG security group instead.

To change the security group of the NAT instance

1. Open the Amazon EC2 console at https://console.aws.amazon.com/ec2/.
2. In the navigation pane, choose **Network Interfaces**.
3. Select the network interface for the NAT instance from the list and choose **Actions**, **Change Security Groups**.
4. In the **Change Security Groups** dialog box, for **Security groups**, select the NATSG security group that you created (see Security) and choose **Save**.